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Internship Report

Phage-Host Classification Using Conformal Prediction????

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Abstract

Acknowledgements

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Contents

1 Introduction

The rapid rise of antibiotic-resistant bacteria is one of the major challenges that global medicine is facing today. To overcome this issue, researchers are going to alternative solutions, one of which is phage therapy. Phage therapy relies on using the naturally occurring viruses to target and destroy the harmful bacteria.

However, To make this therapy a viable alternative, we first need to understand the basic biology of these viruses. Bacteriophages, or phages are viruses that infect bacterial hosts that and are one of the most abundant entities on the planet. Each phage contains a collection of structural and functional proteins that collectively determine the bacterial strains that the phage can infect - this property is called the host range of the phage.

These phages are highly host-specific: a phage which infects one bacterial host may have very little to no effect on another. This particular property of theirs has naturally generated interest in modelling and applications in the context of....... . Therefore a significant problem in the phage biology is the prediction of the bacterial hosts associated with a given phage. Accurate host-predictions can have applications in .....

complete sentence (treatment???)

find applications

The host specificity of phages is one of phage therapy's greatest advantage, but at the same time, it also brings a very significant practical challenge. While it allows us to target specific harmful bacteria without destroying the beneficial ones, it also causes probelms in finding exactly the appropriate phage for a specific infection. Doing this by only experimental screening can be time-consuming and expensive. This motivates the development of reliable computational models that are capable of predicting these phage-host interactions directly from the protein level information.

The existing computational tools, however have a very important limitation: they typically only produce point predictions, which are oftern represented as a scalar confidence score between 0 and 1 with no associated measure of uncertainty. This lack of having an uncertainty quantification is problematic in clinical contexts, where failing to identify a true host could mean missing a life-saving treatment. Consequently, to build trust in automated host prediction, we need a mathematical framework that is capable of quantifying its own predictive uncertainty and provides reliable confidence guarantees.

The primary objective of this report is to address this limitation by developing an uncertainty aware framework for phage-host prediction using **Conformal Prediction**. Conformal prediction provides a distribution-free framework that allows us to transform standard point predictions into prediction sets that are guaranteed to contain the true bacterial host with a user-defined confidence level.

In this project, we build our phage representations by aggregating collections of protein-level functional scores. This process naturally produced high-dimensional, structured score vectors, as different proteins and functions vary in their data availability. These score vectors also ofter contain missing or noisy information. To handle these challenges, we explored geometric conformal prediction methods that can construct reliable prediction sets even when dealing with incomplete biological data. As an intermediate step to validate our geometric approach in a controlled environment, we first evaluated the framework on a binary Gram-type classification task before moving on to the full multi-host prediction problem.

2 Background & State of the Art

In this section, we provide the biological, computational and mathematical background and context for the methods developed in this report. We will begin by introducing bacteriophages and the host-prediction problem in Section 2.1, which motivates the need for computational approaches to find a solution. We will then go over the already existing and state-of-the-art computational techniques in Section 2.2, progressing from the classical sequence similarity techniques to the much nore efficient and modern protein embeddings techniques. We will also look at the limitations that all the currently exisiting methods have in common, for example the absence of rigourous distribution-free uncertainity quantification. . In the remaining part of the section, 2.3 and 2.4 we will introduce the mathematical framework of conformal prediction and the score-based aggregation method of in [1], on which we will build our work.

define uncertainty quantification at somepoint in the report

2.1 Bacteriophages and Host-Prediction Problem

Bacteriophages, commonly referred to as phages, are viruses that specifically infect and replicate within bacteria. They are one of the most abundant biological entities on Earth with an estimated population of 10^{31} in the global biosphere [2], and therefore play a very important role in shaping the microbial communities and evolution. In the recent years the interest in bacteriophages has increased due to the growing threat of antimicrobial resistance and the development of phage therapy as an alternate treatment method to the traditional antibiotics [3].

One of the key characteristics of of phages is its host range, i.e. the set of bacterial hosts that the phage is capable of infecting. The host range of a phage is not a fixed property, it is determined by molecular interactions between the phage and the bacterial surface structures during the cycle of infection [2], [4]. Each phage attaches itself to the bacterial host, injects its viral genetic material which are able to replicate inside the host. Therefore, some of the phages only infect a few hosts (have a small host range) as they can only infect a limited number of bacterial strains while there are other which are capable of infecting a broad range of bacterias.

Identifying which phage infects which host, can have several applications both in environmental and clinical studies. In clinical, the host range helps us determine how suitable a phage is for designing the treatment for phage therapy (an alternate method to antibiotics). In environmental studies it is valuable to know the host being infected to understand the ecological interactions between viruses and microbial communities [4] and evolution in the biosphere.

However, host range has also brings its own obstacles. In phage therapy, we need to find precisely the phage that infects the given bacterial pathogen which is the cause of the infection, and for that we need a fast reliable host prediction method. Traditionally, the method of determining the host range used to depend on laboratory experiments in which candidate phages were tested aginst a number of bacterial hosts; although this method was able to give accurate reliable results but it was extremely expensive and time consuming. Given the rapid increase in the phage genomes generated by modern technology, the number of possible phage-host interacions has increased astronomically, and this leads us to the need of having a compuational prediction method. We need a way which can complement the lab experiment by providing with a shorter list of candidate phages before we move on to expermiental validation [5] or even directly infer the host information directly from a given phage.

We can formulate this host prediction problem over a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the feature space representing phage representations, $X \in \mathcal{X}$ be a random feature vector representing a phage and let $\mathcal{H} = \{h_1, \cdots, h_M\}$ be the finite label space of possible bacterial hosts.

In a single-host classification setting, the underlying data-generating process yields random pairs $(X, H) \sim \mathbb{P}_{X,H}$, where $H \in \mathcal{H}$ represents the true primary host. In a general multi-host setting, the target is a non-empty subset of hosts $H^* \subseteq \mathcal{H}$, i.e., $H^* \in \mathcal{P}(\mathcal{H}) \setminus \{\emptyset\}$, where $\mathcal{P}(\mathcal{H}) = 2^{\mathcal{H}}$ denotes the power set of $\mathcal{H}$.

Our primary objective is to construct a set-valued prediction mapping:

$$C : \mathcal{X} \longrightarrow \mathcal{P}(\mathcal{H})$$

which maps an unlabelled test phage $X_{n+1} \in \mathcal{X}$ to a prediction set $C(X_{n+1}) \subseteq \mathcal{H}$ such that it guarantees that the true host is included in the prediction set with a with a user specified statistical confidence, $1 - \alpha \in (0, 1)$:

$$\mathbb{P}\left(H_{n+1} \in C(X_{n+1})\right) \geq 1 - \alpha$$

where $\alpha \in (0, 1)$ denotes the target error probability.

2.2 Current Computational Approaches and Limitations

The problem of predicting which phage infects which host bacteria has been studied computationally for over a decade and the methods have significantly evolved during this time. Prediction methods have progressed from simple rule based alignemnt to deep representation learning. Here, we will summarize each the primary methodologies of these methods and along with the common limitations they share that motivate us for this work.

2.2.1 Sequence Similarity and Protein Content Methods

The early host prediction tools rely on sequence alignment between a phage and bacterial genomes, assuming that similarity between the two indicates infection compatibility. HostPhinder [6] represents each genome g by its normalised k-mer frequency vector $\mathbf{v}(g) \in \mathbb{R}^{4^k}$ and assigns the predicted host through nearest-neighbour search over a reference database $\mathcal{D}_{\text{ref}} = \{(g_i, y_i)\}_{i=1}^{N}$:

$$\hat{y} = \operatorname*{argmax}_{y \in \mathcal{H}} \max_{g_i : y_i = y} \operatorname{sim}(\mathbf{v}(g_{\text{test}}), \mathbf{v}(g_i)) .$$

WIsH [7] uses a probabilistic approach, it models each candidate host genome as a variable- order Markov chain with parameters θ_h and then selects the host that maximises the log-likelihood of the new test phage sequence $S = (a_1, \ldots, a_L)$:

$$\hat{y} = \operatorname*{argmax}_{h \in \mathcal{H}} \ \ell(S \mid \theta_h), \qquad \ell(S \mid \theta_h) = \sum_{i=k+1}^{L} \log \mathbb{P}(a_i \mid a_{i-k}, \ldots, a_{i-1};\, \theta_h).$$

The protein-content methods overcome the limitation of whole-genome alignment by decomposing the phage into its constituent proteins. RaFAH [8] clusters phage proteins into M protein clusters via MMseqs2 [9], then it represents each phage by a binary membership vector $\mathbf{x} \in \{0,1\}^M$, and trains a random forest to map this profile to a distribution over host genera.

$$f_{\text{RF}} : \{0,1\}^M \longrightarrow \Delta^{|\mathcal{H}|-1}, \quad \text{where } \hat{\mathbf{p}} = f_{\text{RF}}(\mathbf{x}), \quad \sum_{y \in \mathcal{H}} \hat{p}_y = 1$$

This representation is directly relevant to the present work, as our pipeline operates on a related protein-cluster construction, and the data contamination problem discussed in arises from the same source. refer

2.2.2 Protein Language Models and Embedding-Based Methods

The current state of the art techniques use protein language models (pLMs), which are neural architectures trained on millions of protein sequences through self-supervised learning [10, 11]. Given a protein sequence $A = (a_1, \ldots, a_n) \in \mathcal{A}^n$ over the amino acid alphabet $\mathcal{A}$ of size 20, an encoder Φ_{pLM} produces a residue-level embedding matrix $\mathbf{Z} = \Phi_{\text{pLM}}(A) \in \mathbb{R}^{n \times d}$, it is then mean-pooled to a fixed-length sequence representation:

$$\mathbf{z} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{Z}_{i,:} \ \in \ \mathbb{R}^d.$$

The embedding $\mathbf{z}$ encodes biochemical, structural, and evolutionary information without requiring explicit sequence alignment. Building on ProtT5 [10], tools such as EMPATHI [12] and SUBLYME [13] use these representations for phage protein functional annotation and metagenomic lysin discovery respectively. These tools are directly involved in this work: the K-dimensional score vector $\mathbf{s} \in [0,1]^K$ that serves as input to our conformal framework is produced by a pipeline built on these annotations, the precise construction of which is described in . For host prediction, protein embeddings have shown substantially refer improved accuracy over sequence-similarity baselines [14, 15]. This confirms that the embedding space contains sufficient information for reliable host assignment.

2.2.3 Shared Limitation: Absence of Uncertainty Quantification

Despite their empirical success of all the methods seen, they all share a fundamental limitation: they produce point predictions (a single host label $\hat{y}$ or a scalar score $\hat{s}(x,y) \in [0,1]$) without any formal quantification of prediction uncertainty. A score $\hat{s}(x,y) = 0.87$ reflects a relative ranking statistic, not a probability $\mathbb{P}(Y = y \mid X = x)$. And no existing method provides a finite-sample guarantee that the true host is contained in any predicted set with a specified probability.

Existing uncertainty quantification frameworks only offer partial solutions. For example, the Bayesian approaches require strong assumptions on the distribution distributional that are difficult to justify for biological score data, while post-hoc calibration methods such as Platt scaling provide no finite-sample guarantees on unseen test data [16].

These limitations are consequential in phage therapy, where an incorrectly predicted host can directly harm a patient. This motivates the adoption of conformal prediction [17]. This is a distribution-free framework that wraps any existing scoring model in a statistical construction, and produces prediction sets $C(X) \subseteq \mathcal{H}$ with exact finite-sample coverage guarantees while only requiring exchangeability of calibration and test data.

2.3 Conformal Prediction

Conformal prediction, initially developed by Vovk, Gammerman and Shafer [17] and then later made accessible by Angelopoulos and Bates [16], is a framework for constructing prediction sets that have a coverage guarantee. Unlike the methods seen in 2.2 it can provide us with a set of possible outcomes (not a point prediction), have a mathematical guarantee and yet requires no distributional assumptions on the data, except for the exchangibility condition. Here, we will formalize the framework of conformal prediction and analyze its finite-sample coverage properties.

Definition 2.1 (Exchangeability).
A finite sequence of random variables $Z_1, Z_2, \ldots, Z_m$ taking values in a space $\mathcal{Z}$ is exchangeable if, for every permutation σ of the indices $\{1, 2, \ldots, m\}$, the joint distribution of the permuted sequence is identical to that of the original sequence:

$$(Z_1, Z_2, \ldots, Z_m) \stackrel{d}{=} (Z_{\sigma(1)}, Z_{\sigma(2)}, \ldots, Z_{\sigma(m)})$$

where $\stackrel{d}{=}$ denotes equality in distribution.
Or, equivalently for all measurable sets $A_1, A_2, \ldots, A_m \subseteq \mathcal{Z}$ and all permutations $\sigma \in S_m$:

$$\mathbb{P}(Z_1 \in A_1, Z_2 \in A_2, \ldots, Z_m \in A_m) = \mathbb{P}(Z_{\sigma(1)} \in A_1, Z_{\sigma(2)} \in A_2, \ldots, Z_{\sigma(m)} \in A_m)$$

Independent and identically distributed (i.i.d.) seqeunces are always exchangeable, but the converse does not always hold. Thus, exchangeability is considered to be a weaker condition than i.i.d., since it requires that the observations have no special ordering. This condition can easily be met by splitting any dataset randomly [1, 17].

2.3.1 Mathematical Setup and Prediction Set

Let $(X, Y) \in \mathcal{X} \times \mathcal{Y}$ be a random input-label pair, where $\mathcal{X}$ is the feature space that represents our object of interest (in our case it is the phage protein level score vectors) and $\mathcal{Y} = \{1, \cdots, K\}$ is the discrete label space (set of candidate hosts).

We assume that we have a calibration dataset, $D_{\text{cal}} = \{(X_1, Y_1), \cdots, (X_n, Y_n)\}$, of n exchangable data points drawn from an unknown joint distribution, $P_{X,Y}$. Then, for a new unlabelled test input, X_{n+1} our aim is to to construct a prediction set function $C : \mathcal{X} \to 2^{\mathcal{Y}}$ dependening on the data which maps the test input to a subset of candidate labels, i.e., $C(X_{n+1}) \subseteq \mathcal{Y}$, such that the true label falls in in the set with a defined coverage that is:

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

where $\alpha \in (0, 1)$ is the significance level. In our experiments we set $\alpha = 0.1$ so that the guarantee is 90%.

Unlike point predictors, $C(X)$ dynamically reflects model uncertainty. It returns a singleton $|C(X)| = 1$ when confident, a multi-label set $|C(X)| \geq 2$ when uncertain between candidates, or an empty set $|C(X)| = 0$ for out-of-distribution inputs. The efficiency of the model can be quantified by the expected set size $\mathbb{E}[|C(X)|]$, an optimal predictor should acheive a coverage $\geq 1 - \alpha$ with the smallest possible set size.

2.3.2 Split conformal prediction

In split conformal prediction, the available data is divided into a training set and a disjoint calibration set, $D_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^{n}$. The training set is used to fit an underlying scoring model $\hat{f} : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We then define a nonconformity score function $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ based on $\hat{f}$. This score function quantifies how unusual a pair (x, y) is relative to the learned patterns.

Then using the calibration set , $\{(X_i, Y_i)\}_{i=1}^{n}$, we evaluate the nonconformity scores:

$$S_i = S(X_i, Y_i), \quad \forall i = 1, \cdots, n$$

To guranatee that out error rate is less than α we need to compute the conformal calibration qunatile, $\hat{q}$, from these calibration scores

Definition 2.2 (Conformal Calibration Quantile)**.**
The conformal calibration quantile q_α is defined as the empirical $\frac{\lceil (n+1)(1-\alpha) \rceil}{n}$-th quantile of calibration scores $\{S_1, \ldots, S_n\}$:

$$q_\alpha := \text{Quantile}\left(\{S_1, \ldots, S_n\}\,;\, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$$

Equivalently, expressed using the empirical cumulative distribution function:

$$q_\alpha = \inf\left\{q \in \mathbb{R} : \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}(S_i \leq q) \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right\}$$

Then, for a new input, X_{n+1} the conformal prediction set, $C(X_{n+1})$, is constructed by collecting all the candidate labels in $y \in \mathcal{Y}$ whose nonconformity scores do not exceed q_α:

$$C(X_{n+1}) = \left\{y \in \mathcal{Y} \;\middle|\; s(X_{n+1}, y) \leq q_\alpha\right\} \tag{2.3.1}$$

The validity of this construction follows directly from the exchangeability of the calibration and test points. Since $(Z_1, \ldots, Z_{n+1}) = ((X_1, Y_1), \ldots, (X_{n+1}, Y_{n+1}))$ are exchangeable, the joint distribution of $(S_1, \ldots, S_{n+1})$ is invariant under permutations. Therefore, the rank of S_{n+1} among the $n+1$ scores is uniformly distributed on $\{1, \ldots, n+1\}$, giving:

$$\mathbb{P}(S_{n+1} \leq q_\alpha) = \mathbb{P}(\text{rank}(S_{n+1}) \leq \lceil (n+1)(1-\alpha) \rceil) = \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geq 1 - \alpha.$$

Theorem 2.3 formalises this argument.

Theorem 2.3 (Marginal Coverage Guarantee [17, 18])**.** *If $(X_1, Y_1), \ldots, (X_{n+1}, Y_{n+1})$ are exchangeable, then the prediction set $C(X_{n+1})$ defined in Equation (2.3.1) satisfies:*

$$\mathbb{P}\left(Y_{n+1} \in C(X_{n+1})\right) \geq 1 - \alpha$$

Furthermore, if calibration scores $S_1, \ldots, S_n$ are almost surely distinct continuous random variables, the coverage bound is tight:

$$1 - \alpha \leq \mathbb{P}\left(Y_{n+1} \in C(X_{n+1})\right) \leq 1 - \alpha + \frac{1}{n+1} \tag{2.3.2}$$

Crucially, Theorem 2.3 holds strictly in finite samples without requiring parametric assumptions on $s(\cdot,\cdot)$ or $\mathbb{P}_{X,Y}$. While the choice of scoring function s governs efficiency (set size), valid statistical coverage is guaranteed by construction in a single computational pass over D_{cal} [1].

2.3.3 Marginal & Conditional Coverage

Standard conformal prediction satisfies marginal coverage, which means that the $(1-\alpha)\%$ coverage guarantee holds on average across all possible choices of the calibration set, D_{cal}, and test samples, (X_{n+1}, Y_{n+1}):

$$\mathbb{E}_{D_{cal},(X_{n+1},Y_{n+1})}\left[\mathbb{I}(Y_{n+1} \in C(X_{n+1}))\right] = \mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1-\alpha$$

HOwever, the marginal coverage does not ensure the conditional coverage, given specific features $(X = x)$, or specific classes $(Y = y)$. So, for a dataset that has class imbalance, which is very common in phage-host interaction, a marginal model can satisfy the coverage guarantee by potentially over-covering a majority class (100%) and under- covering a small class (0%) [1, 19]. For instance, if $\mathcal{Y} = \{-1, 1\}$ with class proportion $\mathbb{P}(Y = -1) = 0.9$, a predictor acheiving 100% coverage on $Y = -1$ and 0% coverage on $Y = 1$ still attains an overall 90%. In phage-host classification, such conditional under-coverage on minority host may be a problem.

While feature-conditional coverage $(\mathbb{P}(Y \in C(X) \mid X = x) \geq 1-\alpha)$ is mathematically impossible to achieve in finite samples without strong parametric assumptions [17, 1], class-conditional coverage (Mondrian conformal prediction) is exact and valid in finite samples. We partition D_{cal} into K disjoint subsets $D_{\text{cal}}^{(y)} = \{(X_i, Y_i) \in D_{\text{cal}} : Y_i = y\}$ and evaluate separate class-specific quantiles $q_\alpha^{(y)}$, enforcing exact coverage per class:

$$\mathbb{P}\left(Y_{n+1} \in C(X_{n+1}) \,\middle|\, Y_{n+1} = y\right) \geq 1-\alpha, \quad \forall y \in \mathcal{Y} \tag{2.3.3}$$

Because exchangeability holds within each class-conditional partition $D_{\text{cal}}^{(y)}$, evaluating per-class quantiles preserves exact finite-sample validity across every candidate host, as formalised in Section refer

2.4 Conformal Set Aggregation (CSA)

The standard split conformal framework in 2.3.2, takes a scalar non-conformity score $S(x, y) \in \mathbb{R}$. In this report, the prediction pipeline produces a K dimensional score vector of scores $s \in [0,1]^K$ for each phage, where K is the number of distint functional categories. We may loose the geometric structure of the score distribution if we aggregate this vector to a scalar value before applying conformal prediction. For such settings, Ochoa

Rivera, Patel and Tewari [1] have proposed a score based aggregation (CSA) method for conformal prediction.

2.4.1 Ensemble Score Vectors

Let $\mathcal{S} \subseteq [0,1]^K$ be the score space and let $s = (s_1, \cdots, s_K) \in \mathcal{S}$ be the vector of scores produces by the K models, for each phage. Now, the CSA method treats the entire vector as the object of inference rather than reducing it into a single value. The main idea is to define a region, called an envelope, $E \subseteq [0,1]^K$. This envelope consists of the region where the score vectors of the true hosts for that particular phage are concentrated.

Mathematically, we can say that we apply a label dependent transformation $\phi_y : [0,1]^K \to [0,1]^K$ to the score vector, before actually testing if it is part of the envelope. This transformation needs to be such that the for a phage with the true label y, ϕ_y should be close to the origin in the score space. So, we can write our prediction set as:

$$C(s) = \{y \in \mathcal{Y} : \phi_y(s) \in \hat{t} \cdot E\},$$

where $t \cdot E = \{t \cdot \mathbf{x} : \mathbf{x} \in E\}$ represents linear dilation of E in $\mathbb{R}^K$ and $\hat{t}$ is the global scaling factor that is calibrated on the calibration set 2.3.2. The ϕ_y chosen in this work is defined in .

refer to the correct section

2.4.2 The CSA Framework

The core innovation of the CSA method is in the calibration process. It splits the calibration data into two subsets which have different roles. This is necessary for the exchangeability argument that provides us with the coverage guarantee. If we used the same calibration points to construct both, we would get overfitting which would introduce bias which would invalidate the distruibution free coverage guarantee [1].

(i) **Shape Discovery Set (S_1):** Used to model the spatial distribution of scores and learn the geometric shape, orientation, and boundaries of the envelope $E \subset \mathbb{R}^K$. S_1 determines the shape of the region without fixing its scale.

(ii) **Size Scaling Set (S_2):** Used to determine the scalar scaling parameter $\hat{t} \in \mathbb{R}^+$. $\hat{t}$ inflates or deflates E until exact empirical coverage $1-\alpha$ is attained over S_2:

$$\hat{t} = \inf \left\{ t > 0 \;\middle|\; \frac{1}{|S_2|} \sum_{\mathbf{s} \in S_2} \mathbf{1}\left(\phi_Y(\mathbf{s}) \in t \cdot E\right) \geq \frac{\lceil (|S_2|+1)(1-\alpha) \rceil}{|S_2|} \right\} \tag{2.4.1}$$

This $\hat{t}$ is the smallest scaling factor such that the scaled envelope $\hat{t} \cdot E$ achieves the required

$1-\alpha$ coverage. The scaled envelope is defined linearly as:

$$\hat{t} \cdot E = \{t \cdot x : x \in E\}, \quad \hat{t} > 0$$

Finally, for each calibration point $\mathbf{s} \in S_2$, we define the minimal scaling factor required to include it in the envelope as:

$$\tau(\mathbf{s}) = \inf\{t > 0 : \phi_y(\mathbf{s}) \in t \cdot E\}$$

The global scaling factor is then the $(1-\alpha)$ quantile of these values over S_2:

$$\hat{t} = \tau_{\lceil (|S_2|+1)(1-\alpha) \rceil}$$

where $\tau_{(1)} \leq \cdots \leq \tau_{(|S_2|)}$ are the sorted τ-values. By the same exchangeability argument as Theorem 2.3, this construction guarantees $\mathbb{P}(Y_{n+1} \in C(\mathbf{s})) \geq 1-\alpha$.
Splitting calibration into disjoint sets S_1 and S_2 prevents overfitting and ensures that evaluating $\hat{t}$ on S_2 yields a valid distribution-free coverage guarantee [1]. For a test phage with score vector $\mathbf{s}_{n+1}$, candidate label y is included in the prediction set $C(\mathbf{s}_{n+1})$ if and only if $\phi_y(\mathbf{s}_{n+1}) \in \hat{t} \cdot E$.

2.4.3 Limitations of CSA in Classification

This CSA framework was originally developed and analyzed in a regression setting [1], and has some major limitations when it is applied to classification problems:

(i) **Convexity of Envelope:** In [1] it is assumed that the conformal envelope, E, is an intersection of half spaces and therefore is a convex hull, But in classification, the score vectors for different classes concentrate in seperate clusters near different boundaries of the $[0,1]^K$ score space. Therefore, if we force a convex envelope across the regions, we will create a region that covers a large area of empty score space in between, which would inflate the volume of the envelope. This would lead to an over conservative and uninformative prediction sets, $|C(\mathbf{s})|$, as they would be really large.

(ii) **Calibration Inefficiency:** Originally the method uses an iterative numerical optimication or a binary search to find the optimal scaling factor, $\hat{t}$ over S_2. In cases when the score space is high dimensional, the iterative approach can be really expensive computationally.

(iii) **Lack of Per Class Guarantee:** The scaling calculations in the original method only provide us with a marginal guarantee. This can lead to poor performance on the biological datasets, which are imbalances

These limitations provide us with the oppurtunity to improve efficiency, to acheivee the same covergae but with smaller prediction sets. An this is the main motivation behind this work: designing the non-convex envelopes, developing an analytical solution for $\hat{t}$ and finally implementing the class conditional adjustments to ensure that we get statistical guarantees for the phage-host range prediction.

2.4.4 Limitations of CSA in Bounded Classification Spaces

Because CSA was originally developed for multi-output regression [1], applying it directly to bounded classification score spaces $[0,1]^K$ reveals three major limitations:

(i) **Convexity Efficiency Loss:** Ochoa Rivera et al. assume E is an intersection of half-spaces (a convex hull). In classification, transformed nonconformity vectors concentrate in localized, non-convex directional rays near the origin. Enforcing a convex hull over disconnected clusters forces E to enclose empty void regions $V = \mathrm{Conv}(E) \setminus E$ where no true phage vectors reside, severely inflating set volume and yielding oversized, uninformative prediction sets $|C(\mathbf{s})|$.

(ii) **Iterative Calibration Burden:** Originally, finding $\hat{t}$ over S_2 required iterative binary searches or numerical optimizations, which become computationally prohibitive in high-dimensional feature spaces.

(iii) **Marginal-Only Guarantees:** Global scaling $\hat{t}$ provides only marginal coverage, leading to under-coverage on class-imbalanced biological datasets.

These limitations directly motivate our methodological contributions in Chapter **??**: designing non-convex envelopes (Radial and Strip geometries), deriving closed-form analytical solutions for per-point τ-values to enable one-pass calibration, and integrating class-conditional thresholds to guarantee reliable host-range predictions.

3 Methods

3.1 Problem Formalization

- Input: a phage represented by a score vector in $[0, 1]^K$ where K is the number of functional dimensions
- Define the label space $\mathcal{Y}$ (finite set of classes)
- Goal: construct a prediction set $C(x) \subseteq Y$ with the conformal guarantee
- The one-envelope design: build one single envelope on all training phages under their true-label NC transformation, rather than separate envelopes per class

3.2 Non Convex envelope for classification

- In the regression setting of the original CSA paper, envelopes are typically convex because the score space is unbounded and the data does not cluster into separated groups
- In our classification setting, the score space is bounded in $[0, 1]^K$ and — crucially — true NC vectors for different classes occupy distinct, well-separated regions of this space (example)
- In the classification, produce tighter prediction sets without sacrificing the coverage guarantee
- The guarantee still holds because it depends only on the exchangeability of the data and the quantile construction, not on the shape of the envelope

3.3 Score Transformation

- General formulation: for each class $y \in \mathcal{Y}$, define a transformation ϕ_y:$[0, 1]^K \rightarrow [0, 1]^K$ that maps the raw score vector to an conformal score vector such that a true infector of class y is near the origin
- Property we require: $phi_y(\mathbf{s})$ is small when $\mathbf{s}$ is consistent with label y, large otherwise
- we make a single unified envelope coherent across all classes

3.4 Envelope Constriction

3.4.1 CSA Two-Stage Calibration

- Training vectors are split into two disjoint halves S_1 and S_2 (50/50 random split).
- S_1 (shape discovery): used to learn the geometry of the envelope.
- S_2 (size scaling): used to calibrate the scaling factor $\hat{t}$ that determines the final size of the envelope.

- The two stages must use disjoint data to preserve the exchangeability argument underlying the coverage guarantee.

3.4.2 τ Computation

- Define $\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\}$, the minimum scaling factor needed to include point $\mathbf{s}$ in the scaled envelope E.
- The global scaling factor is then determined by the $(1-\alpha)$ quantile of the τ values over S_2:
$$\hat{t} = \tau_{(k)} \quad \text{where} \quad k = \lceil (|S_2| + 1)(1-\alpha) \rceil$$
where $\tau_{(k)}$ denotes the k-th smallest value of τ in S_2.
- Key contribution: for each of the three envelope geometries, $\tau(\mathbf{s})$ has a closed-form expression computable in a single pass — no iterative search required.
- Computational benefit: the entire calibration step reduces to one forward pass over S_2 followed by a sort.
- Contrast with the original CSA paper which uses iterative binary search.

3.4.3 Radial Method

3.4.4 Strip Method

3.4.5 1D

3.4.6 Class-Specific $\hat{t}$ Calculation

3.4.1 — CSA Two-Stage Calibration

3.5 $\hat{t}$ Calculation

- class conditional (why and how)

4 Data ....

4.1 Data Description

- The two files: protein-level score matrix (1,853,074 proteins × 40 functional score columns, values in [0,1], NaN when protein not annotated with that function) and metadata file (same 1,853,074 rows, columns: accession, host, host_type, split)
- Scale of the dataset: 18,474 unique phages after aggregation
- The split column: integer 0–4 encoding a genome-cluster-based partition — phages in the same genome cluster are always in the same fold, ensuring test phages are genuinely different from training phages at the sequence level
- Class distribution: gram-neg vs gram-pos vs unknown
- The NaN structure: NaN is structural missingness, not noise — a protein that was not annotated with a given function simply has no score for that function, which is different from a score of zero

4.2 Protein to Phage Aggregation

- The raw data is at the protein level but conformal prediction operates at the phage level — aggregation is needed
- NaN-aware mean pooling: for each phage and each functional score column, compute the mean across all proteins of that phage that have a non-NaN value for that column. Proteins with NaN for a given column are excluded from the mean for that column only
- Why not replace NaN with zero before pooling: treating an unannotated protein as having zero evidence would push the phage's aggregated score artificially toward zero, misrepresenting its biology
- After aggregation: result is a matrix of shape (number of phages) × 40, still with NaNs in cells where none of the phage's proteins were annotated with that function

4.2.1 NaN handling after aggregation

- After mean pooling, some phages still have NaN in certain columns -> this happens when none of their proteins were annotated with a particular function
- Strategy: fill remaining NaNs using column-wise medians computed strictly on the training set (never the test set, to prevent data leakage) - The median cap: any training median above 0.6 is capped at 0.5 -> this prevents an imputed value from falsely pushing a phage toward a particular class - any remaining NaNs after median imputation are filled with 0.5 (neutral value)

4.3 Train-Test Split

- Standard split: split 0 is the held-out test set, splits 1–4 form the training set
- The contamination problem: explain the protein cluster (pc) column -> a protein's cluster may include proteins from phages in different folds. The models that produced the scores were trained on protein clusters, so a protein from a training-fold phage may share a cluster with a test-fold protein, introducing subtle leakage
- The masking fix: the file all_random_pc_splits.pkl encodes for each protein and each (host, function) model which fold that protein was assigned to. Setting scores to NaN where the protein was not in the test set of the relevant model, before aggregation, removes the contaminated predictions. The existing NaN-aware mean pooling then handles these naturally -> this is the "missing value" framing

5 Experiments and Results

5.1 Synthetic Data Validation

5.1.1 Puspose and setup

- Why synthetic validation before real data: allows controlled verification that the coverage guarantee holds exactly, with known ground truth distributions
- General setup: simulate score vectors from known distributions, apply all three envelope methods, verify empirical coverage matches theoretical target $1 - \alpha$

5.1.2 2D Experimenats

- Three scenarios tested in 2D (K=2): - Uncorrelated scores: two independent uniform or Gaussian dimensions
- Correlated scores: scores with positive correlation -> tests whether the envelope adapts to the correlation structure
- Asymmetric predictors: one informative dimension and one uninformative dimension -> tests whether the envelope correctly ignores the noisy dimension
- For each scenario: show the envelope shape visually, report empirical coverage across multiple runs, confirm it matches $1 - \alpha$
- Key observation from 2D: the radial and strip envelopes are visibly non-convex and wrap tightly around the data, while collapsed 1D produces a simple interval

5.1.3 Extension to K Dimensions

- Repeat the experiments for $K \in 2, 5, 10, 20$ with a single host
- Show that all three methods maintain coverage $\geq 1 - \alpha$ across all dimensionalities
- Discuss the effect of increasing K on envelope quality: radial method becomes sparser (fewer S1 points per sector), strip method has more dimension pairs to condition on, collapsed 1D is unaffected
- Show that the analytical τ computation gives identical results to what iterative search would give, confirming correctness

5.1.4 Multiple Hosts

- Extend to a simulated multi-class setting with multiple hosts ($|\mathcal{Y}| = 6, K = 5$)
- Each host's true score distribution is a distinct cluster in $[0, 1]^K$
- Verify that the single unified envelope correctly assigns the true host to the prediction set with probability $\geq 1 - \alpha$ - Show 3D visualisation of the envelope for K=3

5.2 Gram Type Classification

5.2.1 Experimental Setup

- Data: split-0 phages only (80% calibration, 20% test), unknown host types excluded
- Parameters: $\alpha = 0.1, M = 200$ directions (radial), M=20 bins (strip), 10 independent runs to estimate variance
- The NC transformation used: keep scores as-is for one class, flip with 1 - s for the other

5.2.2 Results

- table: overall coverage, per-class coverage, average set size, empty set rate, size-2 rate, singleton accuracy std across 10 runs for all three methods
- Bar charts: set size distribution, overall and per-class coverage

5.3 Host Classification

6 Conclusion and Future Work

6.1 Summary.....

6.2 Limitations

- gram-pos coverage -

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